

Hallucination Detection Report

AI Output Verification · Knowledge Graph Analysis

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10

Triples Extracted

4

Verdicts Passed to detection

1

Inconsistencies

6

Rule Violations

VERDICT BREAKDOWN

2

50% of verdicts

Correct

2

50% of verdicts

Fabricated Fact

EXTRACTED TRIPLES

All verifiable triples extracted from the LLM response prior to canonicalization or knowledge-graph checking.

#	Subject	Predicate	Object	Type	Predicate Definition
1	575	HAS_INTEREST_RATE	12.21%	ENTITY_ATTRIBUTE	Indicates the interest rate associated with the subject.
2	580	HAS_INTEREST_RATE	11.91%	ENTITY_ATTRIBUTE	Indicates the interest rate associated with the subject.
3	579	HAS_INTEREST_RATE	12.03%	ENTITY_ATTRIBUTE	Indicates the interest rate associated with the subject.
4	579	HAS_INTEREST_RATE	11.91%	ENTITY_ATTRIBUTE	Indicates the interest rate associated with the subject.
5	890	HAS_INTEREST_RATE	Below 4.36%	ENTITY_ATTRIBUTE	Indicates the interest rate, specified as less than a certain value, associated with the subject.
6	890	HAS_RISK_TIER	Super prime	ENTITY_ATTRIBUTE	Designates the risk tier classification assigned to the subject.
7	890	HAS_RISK_TIER	Ultra prime	ENTITY_ATTRIBUTE	Designates the risk tier classification assigned to the subject.
8	890	HAS_DOWN_PAYMENT	Below 2.0%	ENTITY_ATTRIBUTE	Specifies a down payment amount below a certain percentage associated with the subject.
9	290	HAS_LOAN_TERM	12 months	ENTITY_ATTRIBUTE	Specifies the duration of the loan term associated with the subject.
10	850	HAS_DOWN_PAYMENT	53%	ENTITY_ATTRIBUTE	Specifies the down payment percentage associated with the subject.

CANONICALIZED TRIPLES

Predicates normalized against the database schema. The Confidence column reflects alignment quality.

Subject	Original Predicate	Object	Canonical Predicate	Conf.
575	HAS_INTEREST_RATE	12.21%	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	HIGH
580	HAS_INTEREST_RATE	11.91%	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	HIGH
579	HAS_INTEREST_RATE	12.03%	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	HIGH
579	HAS_INTEREST_RATE	11.91%	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	HIGH

Subject	Original Predicate	Object	Canonical Predicate	Conf.
890	HAS_INTEREST_RATE	Below 4.36%	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	HIGH
890	HAS_RISK_TIER	Super prime	RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL	MEDIUM
890	HAS_RISK_TIER	Ultra prime	RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL	MEDIUM
890	HAS_DOWN_PAYMENT	Below 2.0%	DOWN_PAYMENT_REQUIRED_AS_UPFRONT_BORROWER_CASH_CONTRIBUTION	MEDIUM
290	HAS_LOAN_TERM	12 months	LOAN_TERM_GRANTED_AS_MAXIMUM_REPAYMENT_WINDOW_OFFERED	MEDIUM
850	HAS_DOWN_PAYMENT	53%	DOWN_PAYMENT_REQUIRED_AS_UPFRONT_BORROWER_CASH_CONTRIBUTION	MEDIUM

SYMBOLIC RULE VIOLATIONS

Violations caught by explicit domain constraints before database lookup. Affected triples were excluded from the evidence retrieval step.

c_interest_rate_range · · MEDIUM

890 → INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE → Below 4.36%

Credit score 890 exceeds maximum valid credit score of 850

c_risk_outside_range · · MEDIUM

890 → RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL → Super prime

Credit score 890 is outside the valid range of 300-850

c_risk_outside_range · · MEDIUM

890 → RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL → Ultra prime

Credit score 890 is outside the valid range of 300-850

c_down_payment_range · · MEDIUM

890 → DOWN_PAYMENT_REQUIRED_AS_UPFRONT_BORROWER_CASH_CONTRIBUTION → Below 2.0%

Credit score 890 exceeds maximum valid credit score of 850

c_loan_term_enum · · **MEDIUM**

290 → LOAN_TERM_GRANTED_AS_MAXIMUM_REPAYMENT_WINDOW_OFFERED → 12 months

Credit score 290 is below minimum valid credit score of 300

c_down_payment_range · · **MEDIUM**

850 → DOWN_PAYMENT_REQUIRED_AS_UPFRONT_BORROWER_CASH_CONTRIBUTION → 53%

Down payment 53% exceeds maximum valid value of 50%

LOGICAL INCONSISTENCIES

Contradictions detected between triples within the LLM response, independent of any external knowledge source.

Type	Severity	Explanation	Triples Involved
DIRECT_CONFLICT	HIGH	Subject '579' is assigned two different annual borrowing cost percentages simultaneously.	579 → INTEREST_RATE_CHARGED_AS_A ANNUAL_BORROWING_COST_PERCENTAGE → 12.03% 579 → INTEREST_RATE_CHARGED_AS_A ANNUAL_BORROWING_COST_PERCENTAGE → 11.91%

DETECTED HALLUCINATIONS

Subject	Predicate	Object	Verdict	Match	Violated	Confidence	Explanation
575	INTEREST_RATE_CHARGED_AS_A ANNUAL_BORROWING_COST_PERCENTAGE	12.21%	Correct	SEM ANTIC	—	●●●●● ●●●●● 1.00	The subject, predicate, and object match a DB candidate, with the object matching semantically. Evidence: subject: 575, predicate: INTEREST_RATE_CHARGED_AS_A ANNUAL_BORROWING_COST_PERCENTAGE, object: 12.21
580	INTEREST_RATE_CHARGED_AS_A ANNUAL_BORROWING_COST_PERCENTAGE	11.91%	Correct	SEM ANTIC	—	●●●●● ●●●●● 1.00	The subject, predicate, and object match a DB candidate, with the object matching semantically. Evidence: subject: 580, predicate: INTEREST_RATE_CHARGED_AS_A ANNUAL_BORROWING_COST_PERCENTAGE, object: 11.91
579	INTEREST_RATE_CHARGED_AS_A ANNUAL_BORROWING_COST_PERCENTAGE	12.03%	Fabricated Fact	NONE	—	■ ■ ■ ■ ■ ■ ■ ■ ■ ■ 0.00	Predicate is known but subject and object are not available in DB candidates.
579	INTEREST_RATE_CHARGED_AS_A ANNUAL_BORROWING_COST_PERCENTAGE	11.91%	Fabricated Fact	NONE	—	■ ■ ■ ■ ■ ■ ■ ■ ■ ■ 0.00	Predicate is known but subject and object are not available in DB candidates.

VERDICT LEGEND

Verdict	Description
Correct	All three triple components verified against the knowledge graph.
Rule Violation	One or more components contradict database evidence.
Multi-Rule Violation	Violation spans multiple knowledge-graph triples simultaneously.
Unverifiable	No usable DB evidence found for any component of this triple.
Predicate Mismatch	Subject entity exists, but the stated predicate is not a valid relation.
Fabricated Fact	Triple asserts a relationship absent from the knowledge graph entirely.